

Project Summary

In this project, I worked on a housing price prediction dataset to understand the factors that influence house prices and to build machine learning models for price prediction. I started by loading and exploring the dataset to understand its structure, features, and target variable. During the data cleaning stage, I checked for missing values and duplicate records and found that the dataset was already clean, which made the preprocessing process easier. Since the dataset contained categorical features, I converted them into numerical form using encoding techniques so that they could be used by machine learning algorithms.

To understand the relationship between different features and house prices, I performed correlation analysis. The results showed that all the features contributed to the prediction process, although some features had a stronger impact on price than others. Based on this analysis, I decided to use all available features for model building.

I then trained and evaluated two regression models: Linear Regression and Random Forest Regressor. The Linear Regression model achieved an R^2 score of approximately 0.65, while the Random Forest Regressor achieved an R^2 score of approximately 0.61. Since a higher R^2 score indicates better predictive performance, Linear Regression performed slightly better on this dataset. The error metrics (MAE and RMSE) also supported this observation, showing that the Linear Regression model produced more accurate predictions overall.

To further analyze the data, I created visualizations including the distribution of house prices, a correlation heatmap, and a graph showing the relationship between area and price. These visualizations helped me better understand the dataset and the factors affecting housing prices. Overall, the project demonstrated how data preprocessing, exploratory analysis, visualization, and machine learning techniques can be combined to build an effective housing price prediction system.